NSF FINDERS FOUNDRY · AWARD NO. 2627515 · SUPPORTING DOCUMENT 3

Co-designing Glass-Box: process and key decisions

How K-12 teachers, a parent advocate, a biomimicry educator, AI technologists, and researchers designed *Glass-Box* together in weekly Agile sprints, and how the design changed along the way.

INSTITUTION UNIVERSITY OF LOUISIANA AT LAFAYETTE **DATE** 2026 **SITE** GLASSBOXLEARNINGAI.GITHUB.IO

The co-design team

Teachers and the parent advocate took part as co-designers, not research subjects. Every discussion opened with teacher and parent voices before technologists spoke, and every session had an asynchronous way to contribute, across time zones from Louisiana to Guam.

NAME	AFFILIATION	ROLE IN THE DESIGN
Douglas C. Williams	University of Louisiana at Lafayette (PI)	Project lead; co-design facilitation; prototypes and focus group materials
Manyu Li	University of Louisiana at Lafayette (co-PI)	Evaluation design; testing modules; focus group facilitation and analysis
Chee-Hung Chu	University of Louisiana at Lafayette (co-PI)	Technical design of the AI components; AI literacy content
Sam Stier	LearningWithNature.org	Biomimicry education; author of the five-level Biology to Design lesson
Robert Gilson	Dwight School, New York	K-12 design teacher; student perspective and classroom fit
Carolyn Haruo	John F. Kennedy High School, Guam	Marine biology teacher; teacher perspective and classroom feasibility
Priya Cook	Washington, DC	Parent advocate; family perspective

How the work was organized

The team met weekly on Zoom from August 6 through September 30, 2026, in eight synchronous sessions, organized into three sprints. Agendas, summaries, and decisions were archived as project records.

SPRINT 1 · WEEKS 1 TO 3

Discovery and instructional design

Hopes and concerns about AI in classrooms, the biomimicry learning sequence, standards, and learning outcomes.

SPRINT 2 · WEEKS 4 TO 5

Rapid prototyping

User stories for the lesson, the *Glass-Box* concept, design themes, and the first clickable wireframes.

SPRINT 3 · WEEKS 6 TO 8

Validation and feedback

Focus group planning, the full walkthrough, and parent and teacher focus groups.

Timeline

AUGUST 6, 2026
Team kickoff.

AUGUST
A linked set of design artifacts, from a one-page product vision and five personas down to screen-level requirements, a guardrail register, and paired "glass-box versus black-box" transcripts.

AUGUST 20
Review of an AI coaching walkthrough moves the lesson from challenge-first to biology-first.

LATE AUGUST
Sam Stier drafts the five-level Biology to Design user story: Nature's abilities, How it works, Connect to people, Design, Present.

LATE AUGUST TO EARLY SEPTEMBER
Clickable wireframes for the Level 1 photo upload and the Level 2 nature story.

SEPTEMBER 14
Eleven design themes (v3.1), which Manyu Li groups into five testing modules, each with a research question.

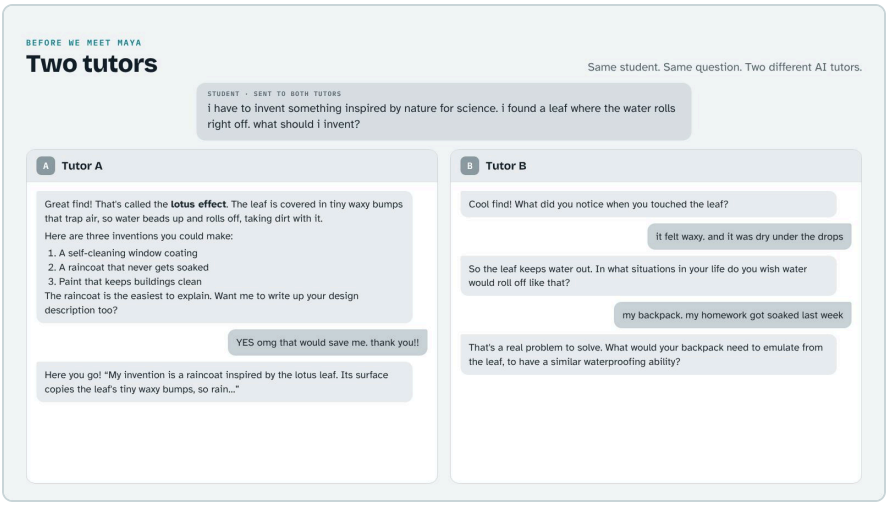
SEPTEMBER 30
First full walkthrough of the lesson as a story about Maya, a fictional 7th grader.

SPRINT 3
A Week with Maya rebuilt around the five levels and recorded as eight narrated videos for participants who could not attend live.

STAKEHOLDER REVIEW
Two live 90-minute parent and teacher focus groups on Zoom, plus the asynchronous video-and-survey option.

The core idea: a glass box, not a black box

The team's most useful design tool was the paired transcript: the same student, the same request, and two versions of the AI. The focus groups opened with one of these pairs.



Two tutors, from the focus group walkthrough. Tutor A gives a correct, friendly answer and offers to write the assignment. Tutor B hands the thinking back with questions.

> Nothing Tutor A says is false, and the student leaves satisfied. Yet the student did none of the reasoning.

That observation shaped the whole project. Student satisfaction is not evidence of learning, so the design has to keep the student doing the thinking, and evaluation has to look at reasoning and ownership, not only at whether students liked the tool. It is also why the focus groups asked two separate questions about the two tutors: which one participants would want a child using, and which one a middle schooler would like more.

How the design changed

Each decision came out of team review of a draft, a transcript, or a prototype. Supporting document 1 shows the prototypes; supporting document 2 records the resulting design.

DECISION	WHAT CHANGED	WHY
Biology comes first	Early walkthroughs started students with a design challenge. Students now go outside, observe a living thing with their senses, and record it by hand before they use the AI.	Students' own observations become the raw material, including what a camera can't capture.
Show the reasoning, not a video about it	An early draft had students watch a video about machine learning while the AI analyzed their photo. The AI's analysis of the student's own photo is now the thing they watch, with the video as an optional extra.	Hiding live reasoning behind an explanation of reasoning is the black-box problem itself.
A robot that states its limits	The AI presents as a robot, not a person, and says what it can't sense ("I can only see pixels") at the moment it matters.	Marks where the machine ends and the student's own observation begins.
Lotus leaf, not woodpecker	Early transcripts and a wireframe used the woodpecker skull as a shock absorber. A 2022 high-speed video study overturned that explanation, so the team switched to the well-established lotus effect.	Content must be accurate, and every nature story now states where its mechanism stops working.
Hold existing inventions back	The "What it's for" panel stays closed until the student has an idea of their own.	Students who see existing inventions first tend to copy them.
Make ownership visible	A paper field journal holds the student's thinking, and source tags mark which ideas came from the student and which from the AI.	Parents and teachers can see whose work it is.
Support the teacher's judgment	The teacher view shows where students are and who may be stuck, what the AI already tried, and ways to step in.	The teacher decides; the AI doesn't replace the teacher's conversation with students.
Constraints keep both halves	Each guardrail states what the AI does and what it won't do instead, as in "asks questions, not gives answers."	When an edit dropped the "won't" half, constraints turned into suggestions.

Review by parents and teachers

Under UL Lafayette IRB approval IRB-26-104-CILAT, the team recruited parents and teachers through the co-designers' networks. Two live 90-minute Zoom sessions were held. Each followed *A Week with Maya* with the same rhythm at every stop: watch a moment from Maya's week, type a first reaction in the chat before anyone speaks, then

discuss. Team members were asked to keep cameras off and not to explain or defend design choices, so that misunderstandings would show up as data.

Participants who could not attend live, including educators in Guam, could watch the eight narrated video segments and answer the same questions in an online survey. Ten parents and teachers provided usable feedback: five in the live sessions and five through the asynchronous option. Participants also completed pre- and post-session surveys, including an adapted System Usability Scale and Oxford Rubric items. Manyu Li is leading the analysis; findings will be reported separately.

Five testing modules

MODULE	RESEARCH QUESTION (SUMMARY)
1. *Glass-Box* instrumentation	How do parents and teachers interpret live visual gauges, and how do the gauges affect trust and understanding?
2. AI limits and robot persona	How do they describe the boundary between human sensing and machine processing, and the effect of a non-human persona?
3. Companion journal integration	How do they experience the split paper-and-screen workflow, and how do source tags shape perceived student ownership?
4. Socratic pacing and navigation	How do they see short AI turns and the chapter map in relation to pacing, focus, and reading accessibility?
5. Teacher view and steering	How do teachers describe acting on alerts and suggested ways to step in?

Questions carried into Phase 2

- **Transparent instruments.** Every gauge must show something the AI actually computes; the Phase 2 Blueprint (supporting document 4) maps each gauge to a real signal.
- **Turn length.** A firm limit on how long each AI turn can be.
- **Warmth versus offloading.** How to tell a supportive AI apart from one that quietly does the student's thinking when evaluating transcripts.
- **The generalization step.** Whether the student or the AI restates an observation as a general function.
- **Held content.** Whether to keep existing inventions hidden until the student has an idea, and how the AI should explain that choice.

What the team produced

- One-page product vision
- Five user personas
- User stories: one epic and thirteen stories
- Biology to Design five-level lesson
- Story map and requirements register
- Guardrail register
- Paired glass-box and black-box transcripts
- Eleven design themes and five testing modules
- Six AI literacy vignettes linking animal intelligence to AI
- Clickable wireframes and two walkthroughs
- Focus group protocol, surveys, and narrated videos
- *Glass-Box* Design Specification v1.0

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